Introduction to Medical Physiology, Definitions, relation of physiology with medicine

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Overview

- Introduction of physiology and human physiology.
- Organization of the human body and properties
- the important life processes of the human body
- Regulation of internal environment

Introduction:

- **Physiology** is the science that seeks to explain the physical and chemical mechanisms that are responsible for the origin, development, and progression of life. is the science of body functions

Human Physiology.

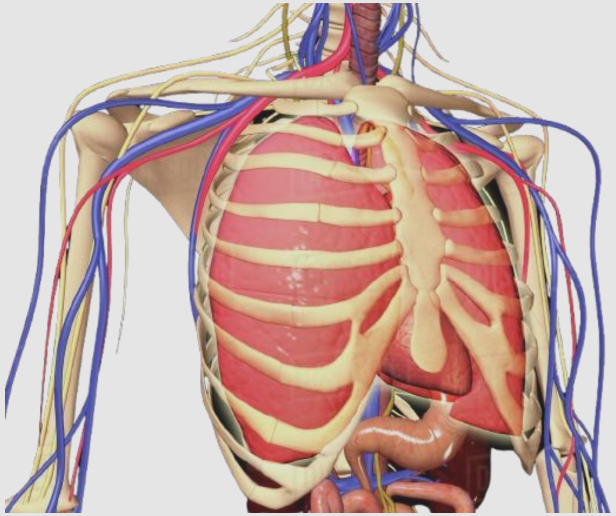
- The science of human physiology attempts to explain the specific characteristics and mechanisms of the human body that make it a living being.
- Human physiology links the basic sciences with medicine and integrates multiple functions of the cells, tissues, and organs into the functions of the living human being.

Structure and Function

- Because structure and function are so closely related, you will learn about the human body by studying its anatomy and physiology together.
- The structure of a part of the body often reflects its functions.
- For example,

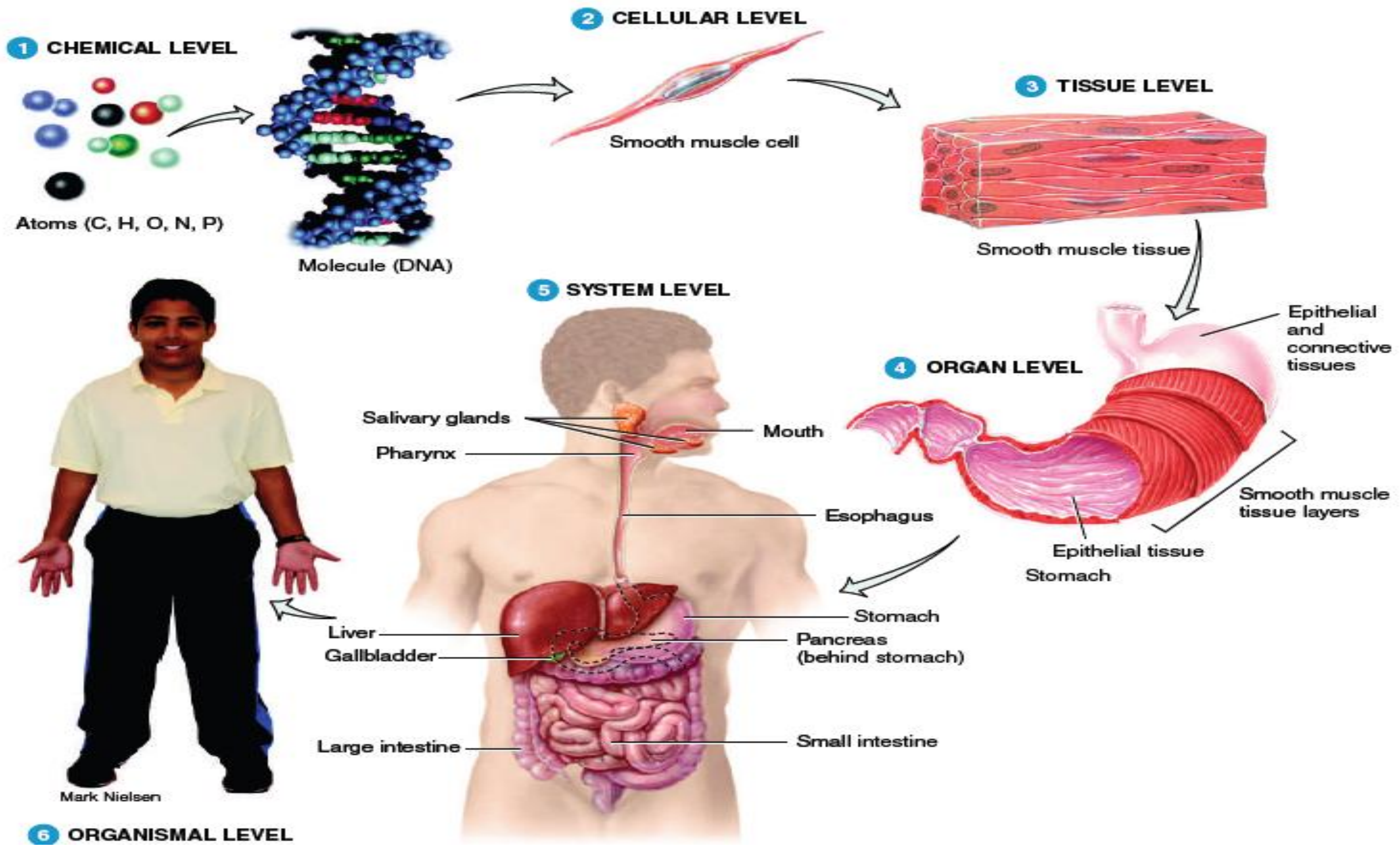
---The bones of the skull join tightly to form a rigid case that protects the brain while the bones of the fingers are more loosely joined to allow a variety of movements.

---The walls of the pulmonary alveoli (air sacs) in the lungs are very thin, permitting rapid movement of inhaled oxygen into the blood.



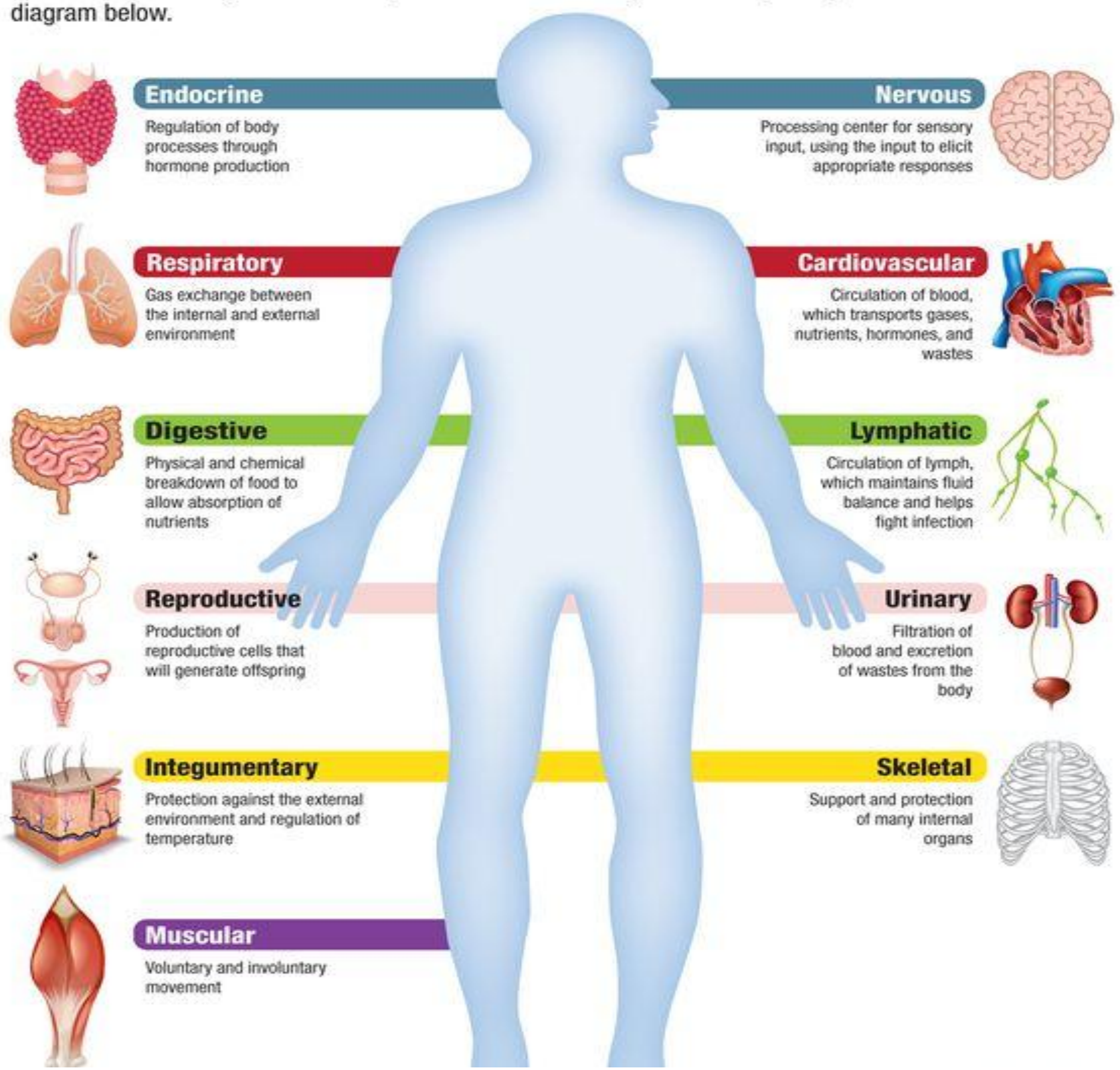
Branch of Physiology	Study of
Molecular physiology	Functions of individual molecules such as proteins and DNA.
Neurophysiology (NOOR-ō-fiz-ē-ol′-ō-jē; *neuro-* = nerve)	Functional properties of nerve cells.
Endocrinology (en′-dō-kri-NOL-ō-jē; *endo-* = within; *-crin* = secretion)	Hormones (chemical regulators in the blood) and how they control body functions.
Cardiovascular physiology (kar-dē-ō-VAS-kū-lar; *cardi-* = heart; *vascular* = blood vessels)	Functions of the heart and blood vessels.
Immunology (im′-ū-NOL-ō-jē; *immun-* = not susceptible)	The body's defenses against disease-causing agents.
Respiratory physiology (RES-pi-ra-tōr-ē; *respira-* = to breathe)	Functions of the air passageways and lungs.
Renal physiology (RĒ-nal; *ren-* = kidney)	Functions of the kidneys.
Exercise physiology	Changes in cell and organ functions due to muscular activity.
Pathophysiology (Path-ō-fiz-ē-ol′-ō-jē)	Functional changes associated with disease and aging.

Levels of Structural Organization



Human Body Systems

There are 11 main systems that keep our bodies functioning. Learn the primary roles of each in the diagram below.



the important life processes of the human body:

Certain processes distinguish organisms, or living things, from nonliving things;

Following are the six most important life processes of the human body:

1. Metabolism:

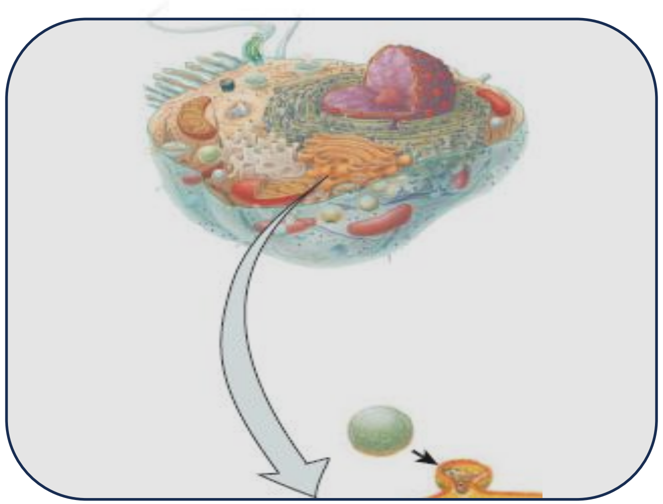
- is the sum of all chemical processes that occur in the body.
- One phase of metabolism is **catabolism** (the breakdown of complex chemical substances into simpler components.
- The other phase of metabolism is **anabolism**, the building up of complex chemical substances from smaller, simpler components.
- For example, digestive processes catabolize (split) proteins in food into amino acids. These amino acids are then used to anabolize (build) new proteins that make up body structures such as muscles and bones.

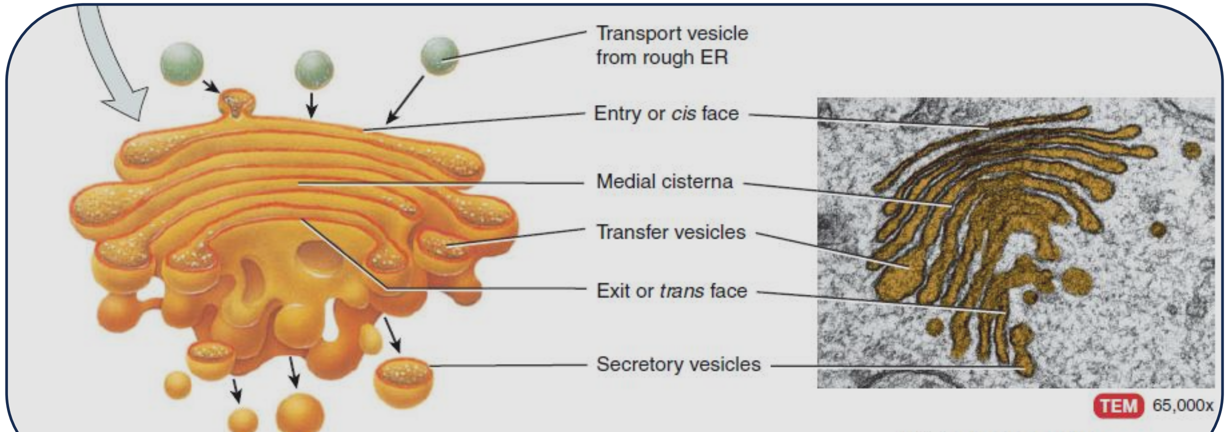
2. Responsiveness

- is the body’s ability to detect and respond to changes. For example, an increase in body temperature during a fever represents a change in the internal environment (within the body), and turning your head toward the sound of squealing brakes is a response to a change in the external environment (outside the body) to prepare the body for a potential threat.
- Different cells in the body respond to environmental changes in characteristic ways.
- Nerve cells respond by generating electrical signals known as nerve impulses (action potentials).
- Muscle cells respond by contracting, which generates force to move body parts.

3. Movement

- includes motion of the whole body, individual organs, single cells, and even tiny structures inside cells.
- For example, the coordinated action of leg muscles moves your whole body from one place to another when you walk or run.
- After you eat a meal that contains fats, your gallbladder contracts and releases bile into the gastrointestinal tract to help digest them.
- When a body tissue is damaged or infected, certain white blood cells move from the bloodstream into the affected tissue to help clean up and repair the area.
- Inside the cell, various parts, such as secretory vesicles, move from one position to another to carry out their functions





(b) Transverse section

4. Growth

- is an increase in body size that results from an increase in the size of existing cells, an increase in the number of cells, or both.
- In addition, a tissue sometimes increases in size because the amount of material between cells increases.
- In a growing bone, for example, mineral deposits accumulate between bone cells, causing the bone to grow in length and width.

5.Differentiation

- Is the development of a cell from an unspecialized to a specialized state.
- Such precursor cells, which can divide and give rise to cells that undergo differentiation, are known as **stem cells**.
- Each type of cell in the body has a specialized structure or function that differs from that of its precursor (ancestor) cells. For example, red blood cells and several types of white blood cells all arise from the same unspecialized precursor cells in red bone marrow.
- Also through differentiation, a single fertilized human egg (ovum) develops into an embryo, and then into a fetus, an infant, a child, and finally an adult.

6.Reproduction

- Refers either to (1) the formation of new cells for tissue growth, repair, or replacement, or (2) the production of a new individual. The formation of new cells occurs through cell division. The production of a new individual occurs through the fertilization of an ovum by a sperm cell to form a zygote, followed by repeated cell divisions and the differentiation of these cells.
- When any one of the life processes ceases to occur properly, the result is death of cells and tissues, which may lead to death of the organism. Clinically, loss of the heartbeat, absence of spontaneous breathing, and loss of brain functions indicate death in the human body.

Homeostasis

A condition of **equilibrium** (balance) in the body's internal environment

- Dynamic condition
- Narrow range is compatible with maintaining life

Example:

- Blood glucose levels range between 70 and 110 mg of glucose/dL of blood
- Whole body contributes to maintain the internal environment within normal limits

Control of Homeostasis

- Homeostasis is constantly being disrupted
  - Physical insults
    - Intense heat or lack of oxygen
  - Changes in the internal environment
    - Drop in blood glucose due to lack of food
  - Physiological stress
    - Demands of work or school

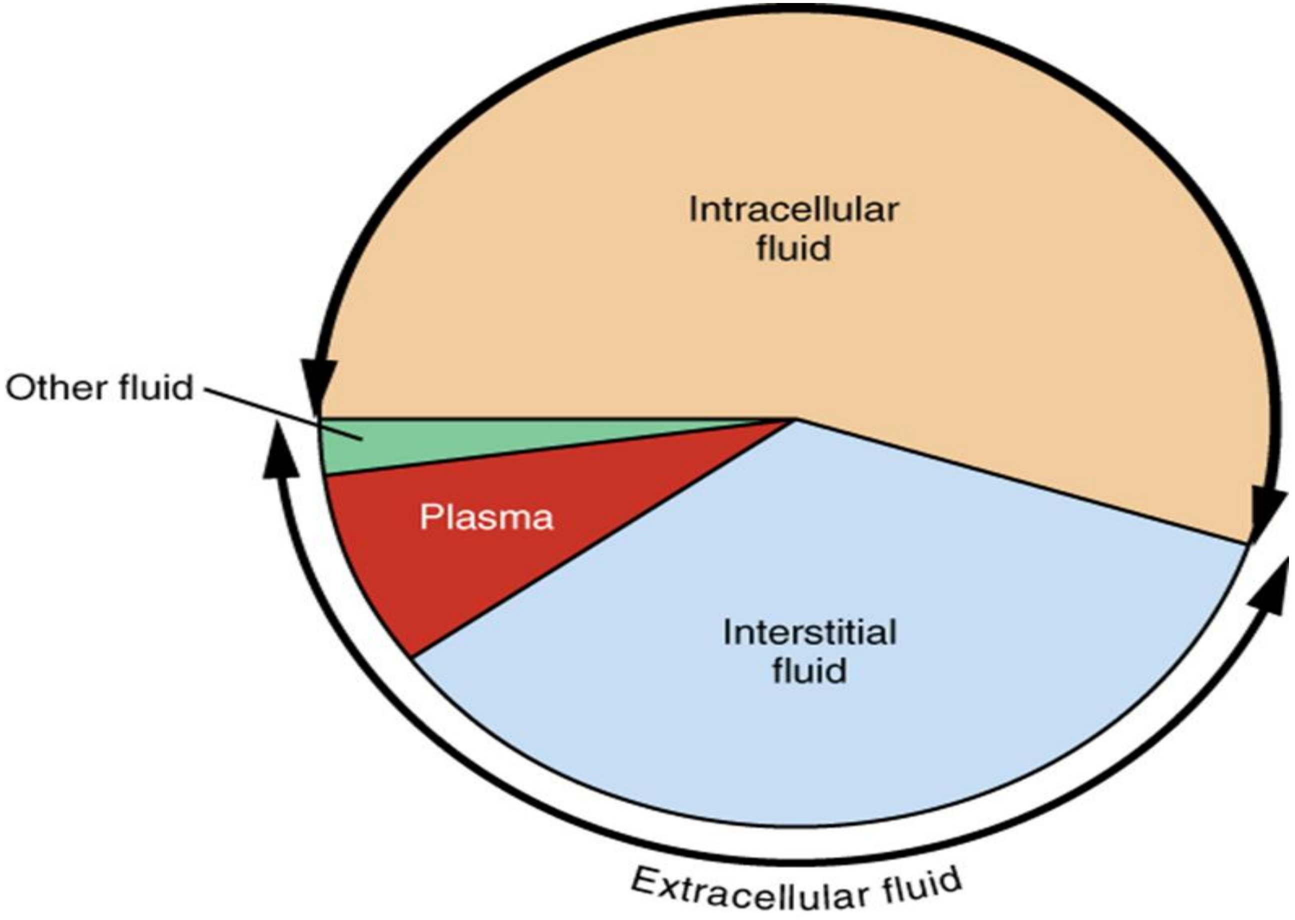
Disruptions are either:

- ➢ Mild and temporary (balance is quickly restored)
- ➢ Intense and Prolonged (poisoning or severe infections)

Homeostasis and Body Fluids

- Maintaining the volume and composition of body fluids are important
  - **Body fluids** are defined as dilute, watery solutions containing dissolved chemicals inside or outside of the cell
  - **Intracellular Fluid (ICF)**
    - Fluid within cells
  - **Extracellular Fluid (ECF)**
    - Fluid outside cells
    - <u>**Interstitial fluid**</u> is ECF between cells and tissues

Most of the water in the body is intracellular fluid. The second largest volume is the interstitial fluid.



A Pie Graph Showing the Proportion of Total Body Fluid in Each of the Body's Fluid Compartments:

Interstitial Fluid and Body Function

- Cellular function depends on the regulation of composition of interstitial fluid
- Body's internal environment
- Composition of interstitial fluid changes as it moves.
- Movement back and forth across capillary walls provide nutrients (glucose, oxygen, ions)to tissue cells and removes waste (carbon dioxide)

ECF and Body Location

- **Blood Plasma**
  - ECF within blood vessels
- **Lymph**
  - ECF within lymphatic vessels
- **Cerebrospinal fluid (CSF)**
  - ECF in the brain and spinal cord
- **Synovial fluid**
  - ECF in joints
- **Aqueous humor and vitreous body**
  - ECF in eyes


Questions?